

1. Footprinting Tools - Conducting Competitive Intelligence

Objective :-

To collect publicly available information about ABC Technologies without directly interacting with its servers.

Tools Used and findings

Information Required	Tool Used	Purpose
Domain Registration	WHOIS	Retrieves domain ownership and registration information
Employee Email Addresses	theHarvester	Collects publicly available email addresses from search engines
Technology stack	Builtwith/ Wappalizer	Identifies technologies used on websites.
publicly exposed sub-domains	Sublist3r/ crtsh	Discovers subdomains associated with the target domain.

1. Domain Registration Details

Tools WHOIS

Command :

whois abctechnologies.com

Information Obtained :

- Domain Registrar
- Registration Date
- Expiration Date
- Name Servers
- Registrant Details

2. Employee Email Addresses

Tool : theHarvester

Command :

theHarvester -d abctechologies.com -b google

Information Obtained :

- Employee email addresses
- Employee name
- Associated hosts

3. Technology Stack Identification

Tool : BuiltWith / Wappalyzer

Information obtained :

- Web Server (Apache, Nginx)
- CMS (Word Press, Drupal)
- JavaScript Framework
- Analytics frameworks/Tools.

4. Publicly Exposed Subdomains

Tool : sublist3r

Command :

sublist3r -d abctechologies.com

Information Obtained:

- mail.abctechnologies.com
- dev.abctechnologies.com
- portal.abctechnologies.com

Role in Competitive Intelligence:

Subdomains reveal additional services and systems exposed to the internet.

Conclusion:-

Passive footprinting helps gather valuable information about a target organization without direct interaction. WHOIS, theHarvester, BuiltWith, and Sublist3r provide insights into domain ownership, employee emails, technologies used, and exposed subdomains.

2) DNS Zone Transfers

Objective:-

To determine whether the target DNS server permits unauthorized DNS transfers.

Tool Used:

dig (Domain information groper)

Command:-

```
dig -AXFR @ns1.targetdomain.com targetdomain.com
```

or

```
host -l targetdomain.com ns1.targetdomain.com
```

Successful Zone Transfer Output

If the DNS server is misconfigured, it may return records

Record type	Example
A Record	web.targetdomain.com
Mx Record	mail.targetdomain.com
NS Record	ns1.targetdomain.com
CNAME Record	vpn.targetdomain.com

Information Revealed

Internal Hosts

Ex: server1.targetdomain.com
dbserver.targetdomain.com

Mail Servers

Ex: mail.targetdomain.com
smtp.targetdomain.com

Subdomains

Ex: vpn.targetdomain.com
dev.targetdomain.com
portal.targetdomain.com

Reconnaissance Value

A successful zone transfer can help an attacker:

1. Map the organization's network.
2. Identify internal and external hosts.
3. Discover email infrastructure.
4. Find additional attack surfaces.

Conclusion :-

DNS zone transfers should only be allowed b/w trusted DNS misconfigured DNS servers can expose valuable reconnaissance information that may aid attackers.

3. Social Engineering Attacks and Countermeasures

Scenario :-

Employees receive an email claiming to be from the IT department requesting credential verification through a provided link because of an urgent security update.

Type of Attack

Phishing Attack

Phishing is a social engineering attack where attackers impersonate a trusted organization to trick users into revealing sensitive information such as usernames and passwords.

Indicators of Phishing

1. Urgent language demanding immediate action.
2. Suspicious or unfamiliar links.
3. Requests for login credentials.
4. Unexpected emails from supposed IT staff.
5. Spelling or grammatical errors.

Countermeasures

1. Verify the sender

check the sender's email address carefully

Benefit: Detects spoofed or fake email accounts.

2. Do Not Click Suspicious Links

Hover over links before clicking.

Benefit: Prevents users from visiting malicious websites.

3. Contact the IT Department Directly

verify the request using official communication channels
Benefit: confirms whether the email is legitimate.

4. Use Multi-factor Authentication (MFA)

Enable MFA on all accounts.

Benefit: Even if passwords are stolen, attackers cannot easily access accounts.

5. Report Suspicious Emails

Forward phishing attempts to the security team.

Benefit: Allows the organisation to block similar attacks

6. Security Awareness Training

Provide regular employee training

Benefit: Improves employee ability to recognize phishing attacks.

How these actions Prevent Credential Theft

- Prevent users from entering credentials on fake websites
- Detect fraudulent emails before any action is taken.
- Add an additional security layer through MFA.
- Reduce the success rate of phishing campaigns.

Conclusion:-

The attack described is a phishing attack designed to steal user credentials. Employees should verify sender info, avoid suspicious links, confirm requests through official channels, enable MFA, and report suspicious emails. These measures significantly reduce the risk.